

Online Inference for Quantile Temporal Difference Learning in Distributional Reinforcement Learning

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Abstract

In this paper, we study how to perform statistical inference for quantile temporal difference learning (QTD) in distributional reinforcement learning. Assuming access to a generative model, we first establish functional central limit theorems for both synchronous and asynchronous QTD, which show that the averaged iterates of QTD converge weakly to a rescaled Brownian motion. We next provide online inference methods. Based on random scaling, the inference procedure constructs an asymptotically pivotal statistic for inference by using the information along the whole QTD path. Meanwhile, the proposed statistic can be computed online without storing the entire trajectory of QTD iterates. This substantially reduces the memory requirement and enables efficient statistical inference in distributional reinforcement learning.

1 Introduction

Distributional reinforcement learning (DRL) [17, 1] extends the classical reinforcement learning framework by modeling the entire distribution of the return rather than only its expectation. This distributional perspective provides a richer characterization of the uncertainty associated with learning agents and has led to a series of successful applications in practice [2, 4]. Since return distributions are generally infinite-dimensional objects, practical DRL methods rely on finite-dimensional approximations. Among these approaches, quantile representations have become one of the most widely used paradigms, originating from Dabney et al. [8], which proposed the quantile temporal difference learning (QTD). By approximating return distributions through a collection

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of quantiles, quantile-based methods provide a flexible representation while retaining important distributional information, leading to algorithms such as quantile regression deep Q-network (QR-DQN) [8] and implicit quantile network (IQN) [7].

Despite the empirical success of QTD and its variants, the statistical understanding of QTD remains limited. In practical reinforcement learning applications, the transition dynamics and reward distributions are unknown, and the learned quantile representations involve statistical uncertainty due to finite samples. Existing theoretical studies of QTD have mainly focused on convergence and consistency results, while its statistical properties, including asymptotic sampling fluctuations and valid uncertainty quantification, remain largely unexplored. This motivates a fundamental question: can QTD support statistically valid estimation and inference for the quantile representation of return distributions?

Answering these questions for QTD is challenging due to the unique structure of quantile projected Bellman operators. In categorical distributional reinforcement learning, the support locations are fixed and the parametrization is described by a finite-dimensional probability vector. As a result, the corresponding categorical-projected Bellman update acts linearly on the probability vectors, which makes it amenable to tools developed for linear stochastic approximation. In contrast, quantile representations parameterize return distributions through inverse distribution functions. Consequently, QTD leads to a nonlinear and non-smooth stochastic approximation recursion, where the stochastic updates are generated by indicator-type transformations of the quantile parameters. Therefore, the resulting non-smooth dependence across different quantile levels prevents a direct application of existing stochastic approximation inference theory and requires a dedicated analysis of the asymptotic behavior of QTD iterates.

In this paper, we develop a statistical inference theory for QTD, a representative quantile-based distributional reinforcement learning algorithm. We establish functional central limit theorems and asymptotic inference procedures for QTD, providing a unified understanding of its statistical behavior.

1.1 Our Contributions

We study distributional policy evaluation with quantile parametrization in a tabular γ -discounted Markov decision process. For a given policy π and quantile level m , let $\boldsymbol{\eta}_m$ denote the unique fixed point of the projected distributional Bellman equation $\boldsymbol{\eta} = \boldsymbol{\Pi}_m \mathcal{T}^\pi \boldsymbol{\eta}$, where $\boldsymbol{\Pi}_m$ is the quantile projection operator and $\mathcal{T}^\pi$ is the distributional Bellman operator. Our goal is to characterize the

asymptotic distribution of quantile temporal-difference learning and to develop valid online inference procedures based on the QTD iterates.

Our main contributions are summarized as follows. Assuming access to a generative model, we establish functional central limit theorems in Theorem 3.1 and 3.2 for the iterates of both synchronous and asynchronous QTD. In particular, after suitable normalization, the partial-sum processes of the iterates converge weakly to rescaled Brownian motions. We also provide explicit characterizations of the corresponding asymptotic covariance matrices. Building on the functional central limit theorems, we develop inference procedures based on random scaling for QTD in Theorem 3.3. We construct an asymptotically pivotal statistic, avoiding the need to explicitly estimate the asymptotic covariance matrices. Moreover, the quantities in the statistic can be updated recursively along the QTD trajectory, so the procedures can be implemented online without storing the entire sequence of iterates.

1.2 Related Works

1.2.1 Distributional Reinforcement Learning.

Distributional reinforcement learning studies the entire distribution of the return rather than only its expectation [1]. Several finite-dimensional representations have been developed, including categorical [1], quantile [8], and generative-model-based representations [10, 9]. Among them, quantile representations have become a widely used paradigm. Starting from quantile temporal-difference learning [8], subsequent works developed more flexible parametrizations, including implicit quantile networks [7] and fully parameterized quantile functions [27].

The theoretical literature on distributional reinforcement learning has mainly focused on convergence and finite-sample guarantees. For categorical representations, Rowland et al. [22] established asymptotic convergence of categorical temporal-difference learning, while Peng et al. [18] derived non-asymptotic convergence rates and sample complexity guarantees. Furthermore, Peng et al. [19] characterized its finite-sample behavior under linear function approximation. For quantile-based methods, Rowland et al. [23] established asymptotic convergence of quantile temporal-difference learning. However, their analysis focuses on consistency and does not characterize the asymptotic distribution of the iterates or provide statistical inference procedures. Separately, Rowland et al. [24] decomposed the estimation error of quantile temporal-difference learning into fixed-point bias and finite-sample variance, showing that it can outperform classical temporal difference learning for

expected return estimation.

1.2.2 Statistical Analysis of Reinforcement Learning.

Statistical inference for reinforcement learning has received increasing attention, with most existing work focusing on value functions. Representative results include high-confidence bounds for off-policy evaluation [26, 13], bootstrap-based inference [11], asymptotic inference for robust value functions [28], confidence intervals for value functions under policy evaluation and policy learning [25, 30], and online inference procedures [16, 15].

Comparatively fewer works study inference for return distributions and other distributional quantities. Chandak et al. [5] and Huang et al. [12] developed procedures for estimating return distribution functions and constructing confidence bands, while Qi et al. [21] studied distributional off-policy evaluation under offline data. Under a generative model, Zhang et al. [29] developed estimation and inference theory for return distributions, and Cheng et al. [6] studied model-based estimation and inference for quantile representations. In contrast, we focus on performing statistical inference for quantile temporal difference learning, a widely used model-free algorithm in quantile distributional reinforcement learning.

The remainder of this paper is organized as follows. In Section 2, we introduce some basic concepts of distributional reinforcement learning. In Section 3, we present our statistical analysis of quantile temporal difference learning. In Section 4, we provide an outlined proof of results in Section 3. We conclude our work in Section 5. Details of the proof are given in the appendices.

2 Preliminaries

In this section, we introduce the necessary background for our work. We review the Markov decision process in Section 2.1, followed by metrics on the space of measures in Section 2.2. We introduce distributional Bellman operator and quantile projection operator in Section 2.3. Finally, in Section 2.5, we present the assumptions under which our main theoretical results are established.

2.1 Problem Setup

We consider a discounted Markov decision process (MDP) specified by the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}_R, P, \gamma \rangle$, where $\mathcal{S}$ and $\mathcal{A}$ are finite state space and finite action space respectively, $\mathcal{P}_R: \mathcal{S} \times \mathcal{A} \rightarrow \Delta([0, 1])$ is the distribution of rewards, $P: \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is the transition probability, and $\gamma \in (0, 1)$ is the

discount factor. Here $\Delta(\cdot)$ denotes the set of probability distributions over some set.

For a fixed policy $\pi: \mathcal{S} \rightarrow \Delta(\mathcal{A})$ and an initial state $S_0 = s \in \mathcal{S}$, a random trajectory $\{(S_t, A_t, R_t)\}_{t=0}^\infty$ can be sampled from the MDP using the following procedure:

$$\begin{aligned} A_t \mid S_t &\sim \pi(\cdot \mid S_t), \\ R_t \mid (S_t, A_t) &\sim \mathcal{P}_R(\cdot \mid S_t, A_t), \\ S_{t+1} \mid (S_t, A_t) &\sim P(\cdot \mid S_t, A_t). \end{aligned}$$

The return of such a trajectory starting from state s is defined as the random variable

$$G^\pi(s) := \sum_{t=0}^{\infty} \gamma^t R_t,$$

which is bounded almost surely by $[0, (1-\gamma)^{-1}]$. The value function $V^\pi(s)$ is defined by the expected return $\mathbb{E}[G^\pi(s)]$. We further denote by $\eta^\pi(s) \in \Delta([0, (1-\gamma)^{-1}])$ the distribution of $G^\pi(s)$, and denote $\boldsymbol{\eta}^\pi = (\eta^\pi(s))_{s \in \mathcal{S}}$ for the collection of return distributions across all states.

2.2 Metrics on the Space of Measures

Denote the space of all probability distributions on $\mathbb{R}$ as $\mathcal{P}$. For $\mu \in \mathcal{P}$, the cumulative distribution function is defined as $F_\mu(x) = \mu(-\infty, x]$, and the quantile function is defined as $F_\mu^{-1}(\tau) = \inf\{x: F_\mu(x) \geq \tau\}$. For $1 \leq p < \infty$ and $\mu, \nu \in \mathcal{P}$, the p -Wasserstein metric between μ and ν is defined as

$$W_p(\mu, \nu) = \left(\int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^p dt \right)^{1/p},$$

and the ∞ -Wasserstein metric is defined as

$$W_\infty(\mu, \nu) = \sup_{t \in [0,1]} |F_\mu^{-1}(t) - F_\nu^{-1}(t)|.$$

Moreover, for any extended metric $d: \mathcal{P} \times \mathcal{P} \rightarrow [0, \infty]$, we can define its supremum extension $\bar{d}: \mathcal{P}^\mathcal{S} \times \mathcal{P}^\mathcal{S} \rightarrow [0, \infty]$ as

$$\bar{d}(\boldsymbol{\eta}, \boldsymbol{\eta}') = \sup_{s \in \mathcal{S}} d(\eta(s), \eta'(s)),$$

which is an extended metric on $\mathcal{P}^\mathcal{S}$.

2.3 Distributional Bellman Operator and Quantile Projection Operator

A fundamental property of the value function is that it satisfies the Bellman equation. Letting $\mathbf{V}^\pi := (V^\pi(s))_{s \in \mathcal{S}}$, we have for any $s \in \mathcal{S}$,

$$\begin{aligned} V^\pi(s) &= [T^\pi \mathbf{V}^\pi](s) \\ &:= \mathbb{E}_{A \sim \pi(\cdot | s), R \sim \mathcal{P}(\cdot | s, A)}[R] + \mathbb{E}_{A \sim \pi(\cdot | s), S' \sim P(\cdot | s, A)}[V^\pi(S')] \\ &= \sum_{a \in \mathcal{A}} \pi(a | s) \int_0^1 r \mathcal{P}_R(dr | s, a) + \sum_{a \in \mathcal{A}, s' \in \mathcal{S}} \pi(a | s) P(s' | s, a) V^\pi(s'). \end{aligned} \tag{1}$$

The operator $T^\pi: \mathbb{R}^{\mathcal{S}} \rightarrow \mathbb{R}^{\mathcal{S}}$ is referred to as the Bellman operator, and Equation (1) characterizes $\mathbf{V}^\pi$ as its unique fixed point.

An analogous relationship holds for the return distributions $\boldsymbol{\eta}^\pi$, known as the distributional Bellman equation. That is, for each $s \in \mathcal{S}$,

$$\begin{aligned} \eta^\pi(s) &= [\mathcal{T}^\pi \boldsymbol{\eta}^\pi](s) \\ &:= \mathbb{E}_{A \sim \pi(\cdot | s), R \sim \mathcal{P}_R(\cdot | s, A), S' \sim P(\cdot | s, A)} \left[(b_{R, \gamma})_{\#} \eta^\pi(S') \right] \\ &= \sum_{a \in \mathcal{A}, s' \in \mathcal{S}} \pi(a | s) P(s' | s, a) \int_0^1 (b_{r, \gamma})_{\#} \eta^\pi(s') \mathcal{P}_R(dr | s, a). \end{aligned}$$

Here $b_{r, \gamma}: \mathbb{R} \rightarrow \mathbb{R}$ denotes the affine map $b_{r, \gamma}(x) = r + \gamma x$, and $g_{\#} \mu$ is the pushforward of a measure μ under g , defined by $g_{\#} \mu(B) = \mu(g^{-1}(B))$ for all Borel sets B . The integral $\int_0^1 (b_{r, \gamma})_{\#} \eta^\pi(s') \mathcal{P}_R(dr | s, a)$ is defined in the sense that for any Borel set B ,

$$\left[\int_0^1 (b_{r, \gamma})_{\#} \eta^\pi(s') \mathcal{P}_R(dr | s, a) \right] (B) = \int_0^1 \left[(b_{r, \gamma})_{\#} \eta^\pi(s') \right] (B) \mathcal{P}_R(dr | s, a).$$

Recall that $\mathcal{P}$ is the space of all probability measures on $\mathbb{R}$, and $\mathcal{T}^\pi: \mathcal{P} \rightarrow \mathcal{P}$ is referred to as the distributional Bellman operator, of which the fixed point is the return distribution $\boldsymbol{\eta}^\pi$.

Because the exact distribution $\boldsymbol{\eta}^\pi$ is infinite-dimensional and cannot be computed exactly, we approximate it using a quantile-parameterized distribution. The space of all quantile-parametrized probability distributions is defined as

$$\mathcal{P}_m := \left\{ \nu_{\mathbf{x}} = \frac{1}{m} \sum_{i=1}^m \delta_{x_i} \mid \mathbf{x} = (x_1, \dots, x_m)^\top \in \mathbb{R}^m, x_1 \leq \dots \leq x_m \right\},$$

which is a mixture of Dirac measures and $m \in \mathbb{N}$. We define the quantile projection operator

$\Pi_m: \mathcal{P} \rightarrow \mathcal{P}_m$ as

$$\Pi_m \nu = \frac{1}{m} \sum_{i=1}^m \delta_{F_\nu^{-1}(\tau_i)},$$

where $\tau_i = \frac{2i-1}{2m}$. We lift Π_m to the product space $\mathcal{P}^{\mathcal{S}}$ by defining $(\mathbf{\Pi}_m \boldsymbol{\eta})(s) := \Pi_m \eta(s)$ for any $\boldsymbol{\eta} = (\eta(s))_{s \in \mathcal{S}} \in \mathcal{P}^{\mathcal{S}}$. The properties of $\mathcal{T}^\pi$ and $\mathbf{\Pi}_m$ are summarized in the following proposition.

Proposition 2.1. [3, Proposition 4.15, Lemma 5.25] *The following statements hold:*

- $\mathcal{T}^\pi$ is γ -contractive under the $\bar{W}_p$ metric for every $p \in [1, \infty]$, namely for every $\boldsymbol{\eta}, \boldsymbol{\eta}' \in \mathcal{P}^{\mathcal{S}}$,

$$\bar{W}_p(\mathcal{T}^\pi \boldsymbol{\eta}, \mathcal{T}^\pi \boldsymbol{\eta}') \leq \gamma \bar{W}_p(\boldsymbol{\eta}, \boldsymbol{\eta}');$$

- $\mathbf{\Pi}_m$ is non-expansive under the $\bar{W}_\infty$ metric, namely $\bar{W}_\infty(\mathbf{\Pi}_m \boldsymbol{\eta}, \mathbf{\Pi}_m \boldsymbol{\eta}') \leq \bar{W}_\infty(\boldsymbol{\eta}, \boldsymbol{\eta}')$ for every $\boldsymbol{\eta}, \boldsymbol{\eta}' \in \mathcal{P}^{\mathcal{S}}$.

It immediately follows from Proposition 2.1 that the quantile projected Bellman operator $\mathbf{\Pi}_m \mathcal{T}^\pi$ is a γ -contraction in the Polish space $(\mathcal{P}^{\mathcal{S}}, \bar{W}_\infty)$. Hence, the quantile projected Bellman equation $\boldsymbol{\eta} = \mathbf{\Pi}_m \mathcal{T}^\pi \boldsymbol{\eta}$ admits a unique solution $\boldsymbol{\eta}_m$. Denote $[m] = \{1, \dots, m\}$, and we define the quantile parameter $\boldsymbol{\theta}_m \in \mathbb{R}^{\mathcal{S} \times [m]}$ by

$$\eta_m(s) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta_m(s,i)},$$

with $\theta_m(s, 1) \leq \dots \leq \theta_m(s, m)$ for every $s \in \mathcal{S}$.

Finally, we need a generalized version of quantile projection operator. For $\boldsymbol{\lambda} \in [0, 1]^{\mathcal{S} \times [m]}$, define $\mathbf{\Pi}_m^\lambda$ by

$$(\mathbf{\Pi}_m^\lambda \boldsymbol{\eta})(s) = \frac{1}{m} \sum_{i=1}^m \delta_{(1-\lambda(s,i))F_{\eta(s)}^{-1}(\tau_i) + \lambda(s,i)\bar{F}_{\eta(s)}^{-1}(\tau_i)},$$

where $\bar{F}_{\eta(s)}^{-1}(\tau) = \inf\{x : F_{\eta(s)}(x) > \tau\}$. It is proved in Rowland et al. [23] that every $\mathbf{\Pi}_m^\lambda$ is non-expansive under the $\bar{W}_\infty$ metric. Therefore, every $\mathbf{\Pi}_m^\lambda \mathcal{T}^\pi$ has a unique fixed point.

2.4 Quantile Temporal Difference Learning

We introduce quantile temporal difference learning (QTD) in this section. We introduce the synchronous case first. In this setting, at every iteration t , for every $s \in \mathcal{S}$, we sample

$$a^{(t,s)} \sim \pi(\cdot | s), \left(r^{(t,s)} s'^{(t,s)} \right) \sim Q_{s,a^{(t,s)}}$$

from a generative model and update

$$\theta^{(t)}(s, i) = \theta^{(t-1)}(s, i) + \alpha_{t-1} \left[\tau_i - \frac{1}{m} \sum_{j=1}^m \mathbb{1} \left\{ r^{(t,s)} + \gamma \theta^{(t-1)}(s'^{(t,s)}, j) < \theta^{(t-1)}(s, i) \right\} \right], \quad (2)$$

where we define

$$Q_{s,a} := \mathcal{P}_R(\cdot \mid s, a) \otimes P(\cdot \mid s, a).$$

In an asynchronous setting, at every iteration t , we sample

$$s^{(t)} \sim \mu(\cdot), a^{(t)} \sim \pi(\cdot \mid s^{(t)}), \left(r^{(t)} s'^{(t)} \right) \sim Q_{s^{(t)}, a^{(t)}}$$

from a generative model and update

$$\theta^{(t)}(s, i) = \theta^{(t-1)}(s, i) + \alpha_{t-1} \mathbb{1} \left\{ s = s^{(t)} \right\} \left[\tau_i - \frac{1}{m} \sum_{j=1}^m \mathbb{1} \left\{ r^{(t)} + \gamma \theta^{(t-1)}(s'^{(t)}, j) < \theta^{(t-1)}(s, i) \right\} \right]. \quad (3)$$

2.5 Main Assumptions

For every $(s, a) \in \mathcal{S} \times \mathcal{A}$, we make the following assumption about $\mathcal{P}_R(\cdot \mid s, a)$.

Assumption 1. *For any $s \in \mathcal{S}$, $a \in \mathcal{A}$, $\mathcal{P}_R(\cdot \mid s, a)$ is supported on $[0, 1]$ and has a Lebesgue density $p_{s,a}$. Moreover, $p_{s,a}$ is Lipschitz continuous on $[0, 1]$ and there exists a positive constant C_0 such that $0 < p_{s,a}(x) \leq C_0$ for any $x \in (0, 1)$.*

Moreover, we make an assumption on the locations of θ_m .

Assumption 2. *For every $s, s' \in \mathcal{S}$ satisfying*

$$P^\pi(s' \mid s) := \sum_{a \in \mathcal{A}} \pi(a \mid s) P(s' \mid s) > 0,$$

we have

$$\theta_m(s, i) - \gamma \theta_m(s', j) \notin \{0, 1\}$$

for every $i, j \in [m]$.

This is a technical condition that excludes boundary-degenerate configurations in which Bellman-shifted quantile locations coincide with the endpoints of the reward support. In particular, it

guarantees that the projected Bellman equation is continuously differentiable around the true parameter, thereby ensuring the regularity conditions required by the Z -estimation theory used in subsequent analysis. We suppose that the two assumptions above hold throughout the paper.

3 Main Results

3.1 Functional Central Limit Theorem for QTD

In this section, we present functional central limit theorems (FCLT) for QTD. Let $\mathbf{B}(\cdot)$ denote an $\mathcal{S} \times [m]$ -dimensional standard Brownian motion on $[0, 1]$. We first need to introduce some matrices in preparation.

For a distribution ν on $\mathcal{S}$, $\mathbf{D}_\nu$ is defined as $\text{diag}_s \{\nu(s) \mathbf{I}_m\}$. $\mathbf{G}$ is defined as

$$\begin{aligned} & \mathbf{G}_{(s,i),(s',j)} \\ &= -\frac{\gamma}{m} \sum_{a \in \mathcal{A}} \pi(a | s) P(s' | s, a) p_{s,a} (\theta_m(s, i) - \gamma \theta_m(s', j)) \\ &+ \frac{\mathbb{1}\{(s, i) = (s', j)\}}{m} \sum_{a \in \mathcal{A}, \tilde{s} \in \mathcal{S}} \sum_{k=1}^m \pi(a | s) P(\tilde{s} | s, a) p_{s,a} (\theta_m(s, i) - \gamma \theta_m(\tilde{s}, k)). \end{aligned}$$

For $s \in \mathcal{S}$ and $i \in [m]$, define

$$\xi_{s,i}(a, r, s') := \frac{1}{m} \sum_{j=1}^m \mathbb{1}\{r + \gamma \theta_m(s', j) < \theta_m(s, i)\} - \tau_i.$$

The covariance matrices $\mathbf{\Gamma}_{\text{syn}}$ and $\mathbf{\Gamma}_{\text{asyn}}$ are defined by

$$(\mathbf{\Gamma}_{\text{syn}})_{(s,i),(\tilde{s},k)} = \mathbb{1}\{s = \tilde{s}\} \text{Cov}_{A \sim \pi(\cdot | s), (R, S') \sim Q_{s,A}} [\xi_{s,i}(A, R, S'), \xi_{\tilde{s},k}(A, R, S')]; \quad (4)$$

$$(\mathbf{\Gamma}_{\text{asyn}})_{(s,i),(\tilde{s},k)} = \text{Cov}_{S \sim \mu, A \sim \pi(\cdot | S), (R, S') \sim Q_{S,A}} [\mathbb{1}\{S = s\} \xi_{s,i}(A, R, S'), \mathbb{1}\{S = \tilde{s}\} \xi_{\tilde{s},k}(A, R, S')]. \quad (5)$$

In particular, we can verify that $\mathbf{\Gamma}_{\text{asyn}} = \mathbf{D}_\mu \mathbf{\Gamma}_{\text{syn}}$. Now we can formulate the FCLT.

Theorem 3.1. *For update rules (2), as $T \rightarrow \infty$,*

$$\frac{1}{\sqrt{T}} \sum_{t=0}^{\lfloor Tu \rfloor} (\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}_m) \Rightarrow \mathbf{G}^{-1} \mathbf{\Gamma}_{\text{syn}}^{\frac{1}{2}} \mathbf{B}(\cdot).$$

Theorem 3.2. *For update rules (3), as $T \rightarrow \infty$,*

$$\frac{1}{\sqrt{T}} \sum_{t=0}^{\lfloor Tu \rfloor} \left(\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}_m \right) \Rightarrow (\mathbf{D}_\mu \mathbf{G})^{-1} \boldsymbol{\Gamma}_{\text{asyn}}^{\frac{1}{2}} \mathbf{B}(\cdot).$$

3.2 Online Inference Procedure for QTD

In this section, we provide an online statistical inference procedure for QTD based on the FCLTs established above. The proposed procedure is based on random scaling and does not require estimating either the Jacobian matrix $\mathbf{G}$ or the covariance matrices $\boldsymbol{\Gamma}_{\text{syn}}$ and $\boldsymbol{\Gamma}_{\text{asyn}}$.

Let

$$\bar{\boldsymbol{\theta}}_T := \frac{1}{T} \sum_{t=1}^T \boldsymbol{\theta}^{(t)}$$

be the Polyak–Ruppert averaged QTD iterate. Define the random scaling matrix

$$\hat{\mathbf{V}}_T := \frac{1}{T^3} \sum_{t=1}^T \left(\sum_{k=1}^t \boldsymbol{\theta}^{(k)} - t \bar{\boldsymbol{\theta}}_T \right) \left(\sum_{k=1}^t \boldsymbol{\theta}^{(k)} - t \bar{\boldsymbol{\theta}}_T \right)^\top. \quad (6)$$

Equivalently, denoting

$$\bar{\boldsymbol{\theta}}_t := \frac{1}{t} \sum_{k=1}^t \boldsymbol{\theta}^{(k)},$$

we have

$$\hat{\mathbf{V}}_T = \frac{1}{T^3} \sum_{t=1}^T t^2 (\bar{\boldsymbol{\theta}}_t - \bar{\boldsymbol{\theta}}_T) (\bar{\boldsymbol{\theta}}_t - \bar{\boldsymbol{\theta}}_T)^\top. \quad (7)$$

Let $B(\cdot)$ denote a one-dimensional standard Brownian motion on $[0, 1]$ and define the random variable

$$\mathcal{V} := \frac{B(1)}{\left[\int_0^1 (B(u) - uB(1))^2 du \right]^{1/2}}, \quad (8)$$

which is pivotal.

The following theorem provides a random-scaling inference procedure for linear functionals of the QTD fixed point. In particular, the limiting distribution is the same for synchronous and asynchronous QTD and is independent of the corresponding asymptotic covariance matrix. It is proved in Appendix A.

Theorem 3.3. *Let $\mathbf{c} \in \mathbb{R}^{S \times [m]}$ and suppose that*

$$\mathbf{c}^\top \mathbf{G}^{-1} \boldsymbol{\Gamma}_{\text{syn}} \mathbf{G}^{-\top} \mathbf{c} > 0, \quad \mathbf{c}^\top (\mathbf{D}_\mu \mathbf{G})^{-1} \boldsymbol{\Gamma}_{\text{asyn}} (\mathbf{D}_\mu \mathbf{G})^{-\top} \mathbf{c} > 0.$$

Then, as $T \rightarrow \infty$,

$$\frac{\mathbf{c}^\top (\bar{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_m)}{\sqrt{\mathbf{c}^\top \hat{\mathbf{V}}_T \mathbf{c}}} \Rightarrow \mathcal{V}.$$

Let q_α denote the upper α -quantile of the distribution of $\mathcal{V}$ defined in (8). Then an asymptotic confidence interval follows immediately.

Corollary 3.1. *Under the conditions of Theorem 3.3, denote*

$$\mathbf{CI}(\alpha) = \left[\mathbf{c}^\top \bar{\boldsymbol{\theta}}_T - q_{\alpha/2} \sqrt{\mathbf{c}^\top \hat{\mathbf{V}}_T \mathbf{c}}, \mathbf{c}^\top \bar{\boldsymbol{\theta}}_T + q_{\alpha/2} \sqrt{\mathbf{c}^\top \hat{\mathbf{V}}_T \mathbf{c}} \right],$$

where $q_{\alpha/2}$ is the upper $\alpha/2$ -quantile of $\mathcal{V}$. Then,

$$\lim_{T \rightarrow \infty} \mathbb{P}[\mathbf{c}^\top \boldsymbol{\theta}_m \in \mathbf{CI}(\alpha)] = 1 - \alpha.$$

The random scaling matrix can be updated online without storing the entire trajectory. Define

$$\begin{aligned} \mathbf{A}_t &:= \sum_{k=1}^t k^2 \bar{\boldsymbol{\theta}}_k \bar{\boldsymbol{\theta}}_k^\top, \\ \mathbf{b}_t &:= \sum_{k=1}^t k^2 \bar{\boldsymbol{\theta}}_k, \\ q_t &:= \sum_{k=1}^t k^2. \end{aligned}$$

These quantities admit the recursive updates

$$\begin{aligned} \bar{\boldsymbol{\theta}}_t &= \frac{t-1}{t} \bar{\boldsymbol{\theta}}_{t-1} + \frac{1}{t} \boldsymbol{\theta}^{(t)}, \\ \mathbf{A}_t &= \mathbf{A}_{t-1} + t^2 \bar{\boldsymbol{\theta}}_t \bar{\boldsymbol{\theta}}_t^\top, \\ \mathbf{b}_t &= \mathbf{b}_{t-1} + t^2 \bar{\boldsymbol{\theta}}_t, \\ q_t &= q_{t-1} + t^2. \end{aligned}$$

Consequently,

$$\hat{\mathbf{V}}_t = \frac{1}{t^3} (\mathbf{A}_t - \bar{\boldsymbol{\theta}}_t \mathbf{b}_t^\top - \mathbf{b}_t \bar{\boldsymbol{\theta}}_t^\top + q_t \bar{\boldsymbol{\theta}}_t \bar{\boldsymbol{\theta}}_t^\top).$$

Thus, both the averaged QTD iterate and its random scaling matrix can be updated sequentially as new QTD iterates become available.

4 Proof Outlines

In this section, we present proofs of Theorem 3.1 and 3.2.

4.1 Proof of Theorem 3.1 and 3.2

To establish the FCLT for QTD, we first present the following general functional central limit theorem, which is proved in Appendix B.

Theorem 4.1. *Suppose that the $\mathbb{R}^d$ sequence $(\mathbf{Z}_t)_{t \geq 0}$ is defined by*

$$\mathbf{Z}_t = \mathbf{Z}_{t-1} - \alpha_{t-1}[\mathbf{h}(\mathbf{Z}_{t-1}) + \boldsymbol{\epsilon}_{t-1}],$$

where $\mathbf{h}$ is a Borel function with a unique zero $\mathbf{z}^$. Suppose there exists a matrix $\mathbf{M}$ such that there exists $\delta > 0$ satisfying*

$$\|\mathbf{h}(\mathbf{z}) - \mathbf{M}(\mathbf{z} - \mathbf{z}^*)\| \leq L \|\mathbf{z} - \mathbf{z}^*\|^2 \quad (9)$$

for $\|\mathbf{z} - \mathbf{z}^\| \leq \delta$, and $\mathbf{M}$ is a uniformly repulsive matrix (all its eigenvalues have positive real parts). Define $(\mathcal{F}_t := \sigma(\mathbf{Z}_0, \dots, \mathbf{Z}_{t+1}))_{t \geq -1}$ and suppose that*

(i) $\boldsymbol{\epsilon}_t$ is $\mathcal{F}_t$ -measurable;

(ii) $\mathbb{E}(\boldsymbol{\epsilon}_t \mid \mathcal{F}_{t-1}) = 0$;

(iii) there exists $L_1 > 0$ such that $\|\boldsymbol{\epsilon}_t\| \leq L_1$ almost surely;

(iv) $\mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_t^\top \mid \mathcal{F}_{t-1}) \rightarrow \boldsymbol{\Gamma}$ almost surely.

Set $\alpha_t = c(t+1)^{-a}$ with $1/2 < a < 1$ and suppose that $\mathbf{Z}_n \rightarrow \mathbf{z}^$ almost surely. Then*

$$\frac{1}{\sqrt{T}} \sum_{t=0}^{\lfloor Tu \rfloor} (\mathbf{Z}_t - \mathbf{z}^*) \Rightarrow \mathbf{M}^{-1} \boldsymbol{\Gamma}^{\frac{1}{2}} \mathbf{B}(\cdot),$$

where $\mathbf{B}(\cdot)$ is a d -dimensional standard Brownian motion.

To apply Theorem 4.1 in QTD settings, we need the following lemmas, the proofs of which are deferred to Appendix B.

Lemma 4.1. *For every $\boldsymbol{\lambda} \in [0, 1]^{\mathcal{S} \times [m]}$, $\boldsymbol{\eta}_m = \boldsymbol{\Pi}_m^\lambda \mathcal{T}^\pi \boldsymbol{\eta}_m$.*

Lemma 4.2. *For update rules (2) and (3), as $T \rightarrow \infty$, $\boldsymbol{\theta}^{(T)} \rightarrow \boldsymbol{\theta}_m$ almost surely.*

Lemma 4.3. *Both $\mathbf{G}$ and $D_\mu \mathbf{G}$ are uniformly repulsive.*

Proof of Theorem 3.1. Define

$$\begin{aligned} (\mathbf{h}(\boldsymbol{\theta}))_{s,i} &= \sum_{j=1}^m \sum_{a \in \mathcal{A}, s' \in \mathcal{S}} \frac{\pi(a | s) P(s' | s, a)}{m} F_{s,a}(\theta(s, i) - \gamma \theta(s', j)) - \tau_i, \\ (\mathbf{H}^{(t)}(\boldsymbol{\theta}))_{s,i} &= \frac{1}{m} \sum_{j=1}^m \mathbb{1} \left\{ r^{(t,s)} + \gamma \theta(s'^{(t,s)}, j) < \theta(s, i) \right\} - \tau_i. \end{aligned}$$

By construction, $\mathbb{E}[\mathbf{H}^{(t)}(\boldsymbol{\theta})] = \mathbf{h}(\boldsymbol{\theta})$. Therefore, the synchronous QTD update can be written as

$$\begin{aligned} \boldsymbol{\theta}^{(t)} &= \boldsymbol{\theta}^{(t-1)} - \alpha_{t-1} \left[\mathbf{h}(\boldsymbol{\theta}^{(t-1)}) + \mathbf{H}^{(t)}(\boldsymbol{\theta}^{(t-1)}) - \mathbf{h}(\boldsymbol{\theta}^{(t-1)}) \right] \\ &:= \boldsymbol{\theta}^{(t-1)} - \alpha_{t-1} \left[\mathbf{h}(\boldsymbol{\theta}^{(t-1)}) + \boldsymbol{\epsilon}_{t-1} \right]. \end{aligned}$$

Since

$$(\mathbf{h}(\boldsymbol{\theta}))_{s,i} = F_{(\mathcal{T}^\pi \boldsymbol{\eta}_\theta)(s)}(\theta(s, i)) - \tau_i,$$

where

$$\boldsymbol{\eta}_\theta(s) = \frac{1}{m} \sum_{i=1}^m \delta_{\theta(s,i)},$$

if $\boldsymbol{\theta}^*$ is a zero of $\mathbf{h}$, there exists $\boldsymbol{\lambda}$ such that $\boldsymbol{\eta}_{\boldsymbol{\theta}^*}$ is the fixed point of $\boldsymbol{\Pi}_m^\lambda$. Therefore, $\mathbf{h}$ has a unique zero $\boldsymbol{\theta}_m$ by Lemma 4.1. Moreover, direct differentiation gives $\nabla \mathbf{h}(\boldsymbol{\theta}_m) = \mathbf{G}$.

By the Lemma 4.3, $\mathbf{G}$ is uniformly repulsive. Furthermore, Assumptions 1 and 2 ensure that $\nabla \mathbf{h}$ is Lipschitz continuous in a neighborhood of $\boldsymbol{\theta}_m$. Since $\mathbf{h}(\boldsymbol{\theta}_m) = 0$, it follows that

$$\|\mathbf{h}(\boldsymbol{\theta}) - \mathbf{G}(\boldsymbol{\theta} - \boldsymbol{\theta}_m)\| \leq L \|\boldsymbol{\theta} - \boldsymbol{\theta}_m\|^2$$

for all $\boldsymbol{\theta}$ in a sufficiently small neighborhood of $\boldsymbol{\theta}_m$, and hence Equation (9) holds.

It remains to verify the noise conditions in Theorem 4.1. Recall that

$$\boldsymbol{\epsilon}_{t-1} = \mathbf{H}^{(t)}(\boldsymbol{\theta}^{(t-1)}) - \mathbf{h}(\boldsymbol{\theta}^{(t-1)}).$$

With

$$\mathcal{F}_t = \sigma \left(\boldsymbol{\theta}^{(0)}, \dots, \boldsymbol{\theta}^{(t+1)} \right),$$

$\boldsymbol{\epsilon}_t$ is $\mathcal{F}_t$ -measurable. Moreover, since the samples used at iteration t are independent of $\mathcal{F}_{t-2}$, we

have $\mathbb{E}[\boldsymbol{\epsilon}_t \mid \mathcal{F}_{t-1}] = 0$. Since each coordinate of $\mathbf{H}^{(t)}(\boldsymbol{\theta})$ is uniformly bounded, the sequence $(\boldsymbol{\epsilon}_t)_{t \geq 0}$ is uniformly bounded almost surely. Finally, by the almost sure convergence $\boldsymbol{\theta}^{(t)} \rightarrow \boldsymbol{\theta}_m$ and the continuity of the corresponding conditional covariance matrix, $\mathbb{E}[\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_t^\top \mid \mathcal{F}_{t-1}] \rightarrow \boldsymbol{\Gamma}_{\text{syn}}$ almost surely. Thus all the conditions of Theorem 4.1 are satisfied with

$$\mathbf{M} = \mathbf{G}, \boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\text{syn}}.$$

Applying Theorem 4.1 completes the proof. $\square$

Proof of Theorem 3.2. Recall the function $\mathbf{h}$ defined in the proof of Theorem 3.1. Define

$$(\mathbf{H}^{(t)}(\boldsymbol{\theta}))_{s,i} = \mathbb{1} \left\{ s = s^{(t)} \right\} \left[\frac{1}{m} \sum_{j=1}^m \mathbb{1} \left\{ r^{(t)} + \gamma \theta(s'^{(t)}, j) < \theta(s, i) \right\} - \tau_i \right].$$

By construction, $\mathbb{E}[\mathbf{H}^{(t)}(\boldsymbol{\theta})] = \mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta})$. Consequently, the asynchronous QTD update can be written as

$$\begin{aligned} \boldsymbol{\theta}^{(t)} &= \boldsymbol{\theta}^{(t-1)} - \alpha_{t-1} \left[\mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta}^{(t-1)}) + \mathbf{H}^{(t)}(\boldsymbol{\theta}^{(t-1)}) - \mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta}^{(t-1)}) \right] \\ &:= \boldsymbol{\theta}^{(t-1)} - \alpha_{t-1} \left[\mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta}^{(t-1)}) + \boldsymbol{\epsilon}_{t-1} \right]. \end{aligned}$$

Since $\mu(s) > 0$ for every $s \in \mathcal{S}$, $\mathbf{D}_\mu$ is invertible. Hence $\mathbf{D}_\mu \mathbf{h}$ has the same unique zero $\boldsymbol{\theta}_m$ as $\mathbf{h}$. Moreover,

$$\nabla (\mathbf{D}_\mu \mathbf{h})(\boldsymbol{\theta}_m) = \mathbf{D}_\mu \nabla \mathbf{h}(\boldsymbol{\theta}_m) = \mathbf{D}_\mu \mathbf{G}.$$

By Lemma 4.3, $\mathbf{D}_\mu \mathbf{G}$ is uniformly repulsive. Furthermore, since $\nabla \mathbf{h}$ is Lipschitz continuous in a neighborhood of $\boldsymbol{\theta}_m$ and $\mathbf{D}_\mu$ is fixed, $\nabla (\mathbf{D}_\mu \mathbf{h})$ is also Lipschitz continuous in a neighborhood of $\boldsymbol{\theta}_m$. Therefore,

$$\|\mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta}) - \mathbf{D}_\mu \mathbf{G}(\boldsymbol{\theta} - \boldsymbol{\theta}_m)\| \leq L \|\boldsymbol{\theta} - \boldsymbol{\theta}_m\|^2$$

for all $\boldsymbol{\theta}$ in a sufficiently small neighborhood of $\boldsymbol{\theta}_m$, and hence Equation (9) holds.

It remains to verify the noise conditions. Recall that

$$\boldsymbol{\epsilon}_{t-1} = \mathbf{H}^{(t)}(\boldsymbol{\theta}^{(t-1)}) - \mathbf{D}_\mu \mathbf{h}(\boldsymbol{\theta}^{(t-1)}).$$

Then $\boldsymbol{\epsilon}_t$ is $\mathcal{F}_t$ -measurable. Since the sample used at iteration t is independent of $\mathcal{F}_{t-2}$, conditional

on the past we have $\mathbb{E}[\epsilon_t \mid \mathcal{F}_{t-1}] = 0$. Again, each coordinate of $\mathbf{H}^{(t)}(\boldsymbol{\theta})$ is uniformly bounded, so $(\epsilon_t)_{t \geq 0}$ is uniformly bounded almost surely. Furthermore, by the almost sure convergence $\boldsymbol{\theta}^{(t)} \rightarrow \boldsymbol{\theta}_m$ and the continuity of the corresponding conditional covariance matrix, $\mathbb{E}[\epsilon_t \epsilon_t^\top \mid \mathcal{F}_{t-1}] \rightarrow \boldsymbol{\Gamma}_{\text{asyn}}$ almost surely. Thus all the conditions of Theorem 4.1 are satisfied with

$$\mathbf{M} = \mathbf{D}_\mu \mathbf{G}, \boldsymbol{\Gamma} = \boldsymbol{\Gamma}_{\text{asyn}}.$$

Applying Theorem 4.1 completes the proof. $\square$

5 Conclusions

In this paper, we studied how to perform statistical inference for QTD in distributional reinforcement learning. We established functional central limit theorems for the averaged iterates of QTD in both synchronous and asynchronous settings, characterizing their weak convergence to a rescaled Brownian motion and providing explicit expressions for the covariance matrices. Based on these results, we developed online inference procedures using random scaling, which construct an asymptotically pivotal statistic without requiring explicit estimation of the asymptotic covariance. Moreover, we showed that the proposed inference procedures can be implemented online by recursively updating the required quantities along the QTD trajectory, avoiding the need to store the entire sequence of iterates.

There are several interesting issues for future work. One future direction is to generalize our results to more general step size schedules. Another promising direction for future work is to go beyond the asymptotic framework and establish non-asymptotic convergence rates for QTD. Such results would provide a more complete understanding of the finite-sample behavior of QTD and may further facilitate statistical estimation and inference in reinforcement learning.

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A Omitted Proofs in Section 3

Proof. We only present the argument for synchronous QTD, since the asynchronous case follows identically. Define

$$\Phi_T(u) := \frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Tu \rfloor} (\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}_m), \quad u \in [0, 1].$$

Then

$$\begin{aligned} \sqrt{T} (\bar{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_m) &= \Phi_T(1), \\ \frac{1}{\sqrt{T}} \left\{ \sum_{k=1}^t \boldsymbol{\theta}^{(k)} - t\bar{\boldsymbol{\theta}}_T \right\} &= \Phi_T(t/T) - \frac{t}{T} \Phi_T(1). \end{aligned}$$

It follows from Theorem 3.1 and the continuous mapping theorem that

$$\begin{aligned} T\hat{\mathbf{V}}_T &= \frac{1}{T} \sum_{t=1}^T \left\{ \Phi_T(t/T) - \frac{t}{T} \Phi_T(1) \right\} \left\{ \Phi_T(t/T) - \frac{t}{T} \Phi_T(1) \right\}^\top \\ &\Rightarrow \mathbf{G}^{-1} \boldsymbol{\Gamma}_{\text{syn}}^{1/2} \left[\int_0^1 (\mathbf{B}(u) - u\mathbf{B}(1)) (\mathbf{B}(u) - u\mathbf{B}(1))^\top \mathrm{d}u \right] (\mathbf{G}^{-1} \boldsymbol{\Gamma}_{\text{syn}}^{1/2})^\top. \end{aligned}$$

Let $\sigma_c^2 := \mathbf{c}^\top \boldsymbol{\Sigma}_{\text{syn}} \mathbf{c}$, then $\sigma_c^{-1} \mathbf{c}^\top \mathbf{G}^{-1} \boldsymbol{\Gamma}_{\text{syn}}^{1/2} \mathbf{B}(\cdot)$ is a one-dimensional standard Brownian motion.

Therefore, another application of the continuous mapping theorem yields

$$\frac{\mathbf{c}^\top (\bar{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_m)}{\sqrt{\mathbf{c}^\top \hat{\mathbf{V}}_T \mathbf{c}}} \Rightarrow \frac{B(1)}{\left[\int_0^1 (B(u) - uB(1))^2 \mathrm{d}u \right]^{1/2}}.$$

This completes the proof. □

B Omitted Proofs in Section 4

Proof of Theorem 4.1. Denote $\boldsymbol{\Delta}_t = \mathbf{Z}_t - \mathbf{z}^*$ and $\boldsymbol{\rho}_t = h(\mathbf{Z}_t) - \mathbf{M}(\mathbf{Z}_t - \mathbf{z}^*)$, then

$$\boldsymbol{\Delta}_t = (\mathbf{I} - \alpha_{t-1} \mathbf{M}) \boldsymbol{\Delta}_{t-1} - \alpha_{t-1} (\boldsymbol{\epsilon}_{t-1} + \boldsymbol{\rho}_{t-1}).$$

Therefore,

$$\boldsymbol{\Delta}_t = \prod_{i=0}^{t-1} (\mathbf{I} - \alpha_i \mathbf{M}) \boldsymbol{\Delta}_0 - \sum_{i=0}^{t-1} \alpha_i \prod_{l=i+1}^{t-1} (\mathbf{I} - \alpha_l \mathbf{M}) (\boldsymbol{\epsilon}_i + \boldsymbol{\rho}_i).$$

Define

$$\mathbf{A}_i^t = \alpha_i \sum_{k=i}^{t-1} \prod_{l=i}^{j-1} (\mathbf{I} - \alpha_l \mathbf{M}),$$

then

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_{t=0}^{\lfloor Tu \rfloor} \Delta_t &= T^{-\frac{1}{2}} \alpha_0^{-1} \mathbf{A}_0^{\lfloor Tu \rfloor + 1} \Delta_0 \\ &\quad - T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \frac{\alpha_i}{\alpha_{i+1}} \mathbf{A}_{i+1}^{\lfloor Tu \rfloor + 1} \rho_i \\ &\quad - T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \frac{\alpha_i}{\alpha_{i+1}} \left(\mathbf{A}_{i+1}^{\lfloor Tu \rfloor + 1} - \mathbf{A}_{i+1}^{T+1} \right) \epsilon_i \\ &\quad - T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \left(\frac{\alpha_i}{\alpha_{i+1}} \mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1} \right) \epsilon_i \\ &\quad - T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \mathbf{G}^{-1} \epsilon_i \\ &:= I_0 + I_1 + I_2 + I_3 + I_4 \end{aligned}$$

By Lemma 1 in Polyak and Juditsky [20], there exists a constant C such that $\|\mathbf{A}_i^t\| \leq C$ for all t, i .

Therefore,

$$\sup_{u \in [0,1]} \|I_0\| \leq T^{-\frac{1}{2}} C \alpha_0^{-1} \|\Delta_0\| \rightarrow 0$$

almost surely. For I_2 , we know that

$$\sup_{u \in [0,1]} \|I_1\| \lesssim T^{-\frac{1}{2}} \sum_{i=0}^T \|\rho_i\| \rightarrow 0$$

almost surely by Lemma B.1. For I_3 , we have

$$\begin{aligned} &\left\| T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \frac{\alpha_i}{\alpha_{i+1}} \left(\mathbf{A}_{i+1}^{\lfloor Tu \rfloor + 1} - \mathbf{A}_{i+1}^{T+1} \right) \epsilon_i \right\| \\ &= \left\| T^{-\frac{1}{2}} \sum_{i=0}^{\lfloor Tu \rfloor} \alpha_i \sum_{\lfloor Tu \rfloor + 1}^T \prod_{l=i+1}^{j-1} (\mathbf{I} - \alpha_l \mathbf{G}) \epsilon_i \right\| \\ &= \left\| \mathbf{A}_{\lfloor Tu \rfloor + 1}^{T+1} \alpha_{\lfloor Tu \rfloor + 1}^{-1} \sum_{i=0}^{\lfloor Tu \rfloor} \alpha_i \prod_{l=i+1}^{\lfloor Tu \rfloor} (\mathbf{I} - \alpha_l \mathbf{G}) \epsilon_i \right\| \end{aligned}$$

$$\lesssim \alpha_{[Tu]+1}^{-1} \left\| \sum_{i=0}^{[Tu]} \alpha_i \prod_{l=i+1}^{[Tu]} (\mathbf{I} - \alpha_l \mathbf{G}) \boldsymbol{\epsilon}_i \right\|.$$

By Lemma 4 in Li et al. [15],

$$\sup_{u \in [0,1]} \|I_2\| \lesssim \sup_{u \in [0,1]} T^{-\frac{1}{2}} \alpha_{[Tu]+1}^{-1} \left\| \sum_{i=0}^{[Tu]} \alpha_i \prod_{l=i+1}^{[Tu]} (\mathbf{I} - \alpha_l \mathbf{G}) \boldsymbol{\epsilon}_i \right\| \rightarrow 0$$

in probability. Moreover, by Lemma 1 in Polyak and Juditsky [20],

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T \|\mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1}\| = 0.$$

Therefore, by Doob's maximal inequality,

$$\begin{aligned} & \mathbb{E} \left[\sup_{u \in [0,1]} \|I_3\| \right]^2 \\ & \lesssim \mathbb{E} \left\| T^{-\frac{1}{2}} \sum_{i=0}^T \left(\frac{\alpha_i}{\alpha_{i+1}} \mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1} \right) \boldsymbol{\epsilon}_i \right\|^2 \\ & = T^{-1} \sum_{i=0}^T \mathbb{E} \left\| \left(\frac{\alpha_i}{\alpha_{i+1}} \mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1} \right) \boldsymbol{\epsilon}_i \right\|^2 \\ & \lesssim T^{-1} \sum_{i=0}^T \left[\|\mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1}\|^2 + \left(\frac{\alpha_i}{\alpha_{i+1}} - 1 \right)^2 \right] \\ & \lesssim T^{-1} \sum_{i=0}^T \left[\|\mathbf{A}_{i+1}^{T+1} - \mathbf{G}^{-1}\| + (i+1)^{-2} \right] \rightarrow 0. \end{aligned}$$

Finally, by Lemma A.3 in Li et al. [14], $I_4 \Rightarrow \mathbf{G}^{-1} \boldsymbol{\Gamma}_{\text{syn}}^{\frac{1}{2}} \mathbf{B}(\cdot)$ and the proof is completed. $\square$

The following lemma is proved in Appendix C.

Lemma B.1. *With probability 1, we have*

$$\sum_{i=0}^{\infty} (i+1)^{-\frac{1}{2}} \|\boldsymbol{\rho}_i\| < \infty.$$

Proof of Lemma 4.1. By Lemma A.2 in Cheng et al. [6], $p_{(\mathcal{T}^{\pi} \boldsymbol{\eta}_m)(s)}(\theta_m(s, i)) > 0$ for every (s, i) and

$p(\mathcal{T}^\pi \boldsymbol{\eta}_m)(s)$ is continuous at every $\theta_m(s, i)$. Therefore, for every $\boldsymbol{\lambda} \in [0, 1]^{\mathcal{S} \times [m]}$,

$$\boldsymbol{\Pi}_m^\lambda \mathcal{T}^\pi \boldsymbol{\eta}_m = \boldsymbol{\Pi}_m \mathcal{T}^\pi \boldsymbol{\eta}_m = \boldsymbol{\eta}_m.$$

Therefore Lemma 4.1 holds. $\square$

Proof of Lemma 4.2. Define $\boldsymbol{\Lambda} = \left\{ \boldsymbol{\theta}_m^\lambda : \boldsymbol{\eta}_{\boldsymbol{\theta}_m^\lambda} = \boldsymbol{\Pi}_m^\lambda \mathcal{T}^\pi \boldsymbol{\eta}_{\boldsymbol{\theta}_m^\lambda}, \boldsymbol{\lambda} \in [0, 1]^{\mathcal{S} \times [m]} \right\}$. By Theorem 8 in Rowland et al. [23], with probability 1,

$$\lim_{t \rightarrow \infty} \inf_{\boldsymbol{\theta}^* \in \boldsymbol{\Lambda}} \left\| \boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^* \right\| = 0.$$

However, Lemma 4.1 implies that $\boldsymbol{\Lambda} = \{\boldsymbol{\theta}_m\}$ and the conclusion follows. $\square$

Proof of Lemma 4.3. By Gershgorin circle theorem, if λ is an eigenvalue of $\mathbf{G}$,

$$\begin{aligned} \operatorname{Re}(\lambda) &\geq \min_{(s,i)} \left\{ \mathbf{G}_{(s,i),(s,i)} - \sum_{(s',j) \neq (s,i)} |\mathbf{G}_{(s,i),(s',j)}| \right\} \\ &= \min_{(s,i)} \left\{ p(\mathcal{T}^\pi \boldsymbol{\eta}_m)(s) (\theta_m(s, i)) \right\} > 0, \end{aligned}$$

where the last inequality follows from Lemma A.2 in Cheng et al. [6]. Similarly, if λ is an eigenvalue of $\mathbf{D}_\mu \mathbf{G}$,

$$\begin{aligned} \operatorname{Re}(\lambda) &\geq \min_{(s,i)} \left\{ (\mathbf{D}_\mu \mathbf{G})_{(s,i),(s,i)} - \sum_{(s',j) \neq (s,i)} |(\mathbf{D}_\mu \mathbf{G})_{(s,i),(s',j)}| \right\} \\ &= \min_{(s,i)} \left\{ \mu(s) p(\mathcal{T}^\pi \boldsymbol{\eta}_m)(s) (\theta_m(s, i)) \right\} > 0 \end{aligned}$$

and the proof is completed. $\square$

C Technical Lemmas

Proof of Lemma B.1. We construct a Lyapunov function first. Since $\mathbf{M}$ is uniform repulsive, there exists a positive definite matrix $\mathbf{Q}$ satisfying $\mathbf{M}\mathbf{Q} + \mathbf{Q}\mathbf{M}^\top = \mathbf{I}$. In fact, the solution is given by

$$\mathbf{Q} = \int_0^\infty e^{-\mathbf{M}t} e^{\mathbf{M}^\top t} dt.$$

Let $0 < p_- \leq p_+ < \infty$ be the smallest and largest eigenvalues of $\mathbf{Q}$. Then

$$p_- \|\mathbf{z}\|^2 \leq V(\mathbf{z}) := \mathbf{z}^\top \mathbf{Q} \mathbf{z} \leq p_+ \|\mathbf{z}\|^2. \quad (10)$$

Choose κ such that $0 < \kappa \leq \min\{\delta, (4p_+L)^{-1}\}$. Then, for $\|\mathbf{z} - \mathbf{z}^*\| \leq \kappa$, denote $\mathbf{x} =$

$$\begin{aligned} & 2(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} \mathbf{h}(\mathbf{z}) \\ &= 2(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} \mathbf{M}(\mathbf{z} - \mathbf{z}^*) + 2(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} [\mathbf{h}(\mathbf{z}) - \mathbf{M}(\mathbf{z} - \mathbf{z}^*)] \\ &= (\mathbf{z} - \mathbf{z}^*)^\top (\mathbf{Q} \mathbf{M} + \mathbf{M}^\top \mathbf{Q}) (\mathbf{z} - \mathbf{z}^*) + 2(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} [\mathbf{h}(\mathbf{z}) - \mathbf{M}(\mathbf{z} - \mathbf{z}^*)] \\ &= \|\mathbf{z} - \mathbf{z}^*\|^2 + 2(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} [\mathbf{h}(\mathbf{z}) - \mathbf{M}(\mathbf{z} - \mathbf{z}^*)] \\ &\geq \|\mathbf{z} - \mathbf{z}^*\|^2 - 2p_+ K_h \|\mathbf{z} - \mathbf{z}^*\|^3 \\ &\geq \frac{1}{2} \|\mathbf{z} - \mathbf{z}^*\|^2, \end{aligned}$$

Also, for $\|\mathbf{z} - \mathbf{z}^*\| \leq \kappa$,

$$\begin{aligned} \|\mathbf{h}(\mathbf{z})\| &\leq \|\mathbf{M}(\mathbf{z} - \mathbf{z}^*)\| + \|\mathbf{h}(\mathbf{z}) - \mathbf{M}(\mathbf{z} - \mathbf{z}^*)\| \\ &\leq (\|\mathbf{M}\| + K_h \delta) \|\mathbf{z} - \mathbf{z}^*\|. \end{aligned}$$

Denote $H = \|\mathbf{M}\| + K_h \delta$ and

$$\begin{aligned} & V(\mathbf{z} - \mathbf{z}^* - \gamma \mathbf{h}(\mathbf{z})) - V(\mathbf{z} - \mathbf{z}^*) \\ &= -2\gamma(\mathbf{z} - \mathbf{z}^*)^\top \mathbf{Q} \mathbf{h}(\mathbf{z}) + \gamma^2 \mathbf{h}(\mathbf{z})^\top \mathbf{Q} \mathbf{h}(\mathbf{z}) \\ &\leq -\frac{\gamma}{2} \|\mathbf{z} - \mathbf{z}^*\|^2 + \gamma^2 p_+ H^2 \|\mathbf{z} - \mathbf{z}^*\|^2. \end{aligned}$$

Since $\alpha_t \rightarrow 0$, there exists t_0 such that $\alpha_t p_+ H^2 \leq \frac{1}{4}$ and $c\alpha_t \leq 1$ for all $t \geq t_0$, where $c := (4p_+)^{-1} > 0$. Thus, for every $t \geq t_0$ and every $\|\mathbf{z} - \mathbf{z}^*\| \leq \kappa$,

$$\begin{aligned} V(\mathbf{z} - \mathbf{z}^* - \gamma_t \mathbf{h}(\mathbf{z})) &\leq V(\mathbf{z} - \mathbf{z}^*) - \frac{\alpha_t}{4} \|\mathbf{z} - \mathbf{z}^*\|^2 \\ &\leq (1 - c\alpha_t) V(\mathbf{z} - \mathbf{z}^*). \end{aligned}$$

Now we prove the desired conclusion. Fix integer $T \geq t_0$ and define the stopping time

$$\tau_T := \inf \{t \geq T : \|\mathbf{Z}_t - \mathbf{z}^*\| > \delta\}, \quad (11)$$

with $\inf \emptyset = \infty$, and the event $G_T := \{\tau_T = \infty\}$.

We have

$$\begin{aligned} & \mathbb{E}[V(\mathbf{Z}_t - \mathbf{z}^*) \mathbf{1}\{\tau_T \geq t\} \mid \mathcal{F}_{t-1}] \\ & \leq \mathbf{1}\{\tau_T \geq t-1\} \mathbb{E}[V(\mathbf{Z}_t - \mathbf{z}^*) \mid \mathcal{F}_{t-1}] \\ & = \mathbf{1}\{\tau_T \geq t-1\} [V(\mathbf{Z}_{t-1} - \mathbf{z}^*) + \alpha_{t-1}^2 \mathbb{E}[\boldsymbol{\epsilon}_t^\top \mathbf{Q} \boldsymbol{\epsilon}_t \mid \mathcal{F}_{t-1}]] \\ & \leq (1 - c\alpha_{t-1})V(\mathbf{Z}_{t-1} - \mathbf{z}^*) \mathbf{1}\{\tau_T \geq t-1\} + C\alpha_{t-1}^2. \end{aligned}$$

For $t \geq T+1$, define $u_t := \mathbb{E}[V(\mathbf{Z}_t - \mathbf{z}^*) \mathbf{1}\{\tau_T \geq t\}]$, then

$$u_t \leq (1 - c\alpha_{t-1})u_{t-1} + C\alpha_{t-1}^2.$$

By Lemma A.12 in Li et al. [14], for $t \geq T$, $u_t \leq C_T \alpha_t$.

Since $G_T \subseteq \{\tau_T \geq t\}$, we know that for $t \geq T$,

$$\mathbb{E}\|\boldsymbol{\rho}_t\| \mathbf{1}_{G_T} \lesssim \mathbb{E}\|\mathbf{Z}_t - \mathbf{z}^*\|^2 \mathbf{1}_{G_T} \leq C'_T \alpha_t.$$

Therefore, we know that

$$\mathbb{E} \sum_{t=T}^{\infty} \frac{\|\boldsymbol{\rho}_t\| \mathbf{1}_{G_T}}{\sqrt{t+1}} \lesssim \sum_{t=T}^{\infty} (t+1)^{-(a+\frac{1}{2})} < \infty.$$

Hence on G_T , we know that

$$\sum_{t=T}^{\infty} \frac{\|\boldsymbol{\rho}_t\|}{\sqrt{t+1}} < \infty$$

and the conclusion holds. However, since $\mathbf{Z}_t \rightarrow \mathbf{z}^*$ almost surely,

$$\mathbb{P}\left(\bigcup_{T=1}^{\infty} G_T\right) = 1.$$

Therefore, we know that with probability 1,

$$\sum_{t=0}^{\infty} \frac{\|\boldsymbol{\rho}_t\|}{\sqrt{t+1}} < \infty. \quad \square$$